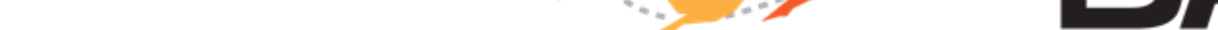


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- Optimize the identification of the most-informative operating conditions
- Develop a dynamic, equation-oriented **model using experimental data**
- Compute model-consistent uncertainty using **intrusive derivatives**
- Design the next experiment to **maximally reduce uncertainty**
- Validate using a membrane separation experiment and generalize

- **Impactful applications**
 - Water treatment
 - Chemical manufacturing
 - Critical-material recovery
- **Membrane systems modeling limitations**
 - **Uncertainty** in transport models
 - Poor parameter **identifiability**
- **Impact on process design**
 - Degraded prediction **accuracy**
 - Unreliable **scalability**

Dynamic Modeling and UQ Workflow

Mechanistic Dynamic Model

$$\begin{aligned} \dot{x}(t) &= f(x(t), u(t), \theta) & u(t) &= \phi \\ \hat{y}_r &= g_r(x, \phi, \theta) & \epsilon_r &\sim N(0, V_r) \\ y_r &= \hat{y}_r + \epsilon_r \end{aligned}$$

Parameter Estimation

$$\hat{\theta} = \arg \min_{\theta} \sum_r (y_r - \hat{y}_r)^T V_r^{-1} (y_r - \hat{y}_r)$$

Intrusive UQ: Covariance/FIM

$$\begin{aligned} \mathbf{Q}_r &= \partial \hat{\mathbf{y}}_r / \partial \boldsymbol{\theta} \Big|_{\hat{\boldsymbol{\theta}}, \phi} \\ \mathbf{M}(\hat{\boldsymbol{\theta}}, \phi) &= \mathbf{V}_{\hat{\boldsymbol{\theta}}}^{-1} \\ \mathbf{V}_{\hat{\boldsymbol{\theta}}} &\approx \left(\sum_r \mathbf{Q}_r^T \mathbf{V}_r^{-1} \mathbf{Q}_r \right)^{-1} \end{aligned}$$

Operating Condition: **MBDoE**

$$\begin{aligned} \phi^* &= \arg \max_{\phi \in \Phi} \psi(M(\widehat{\theta}, \phi)) \\ \psi &\in \{A, D, E, ME\} \end{aligned}$$

Stopping Criterion

$$tr(M(\hat{\theta}, \phi)_k^{-1}) \leq \tau$$

Takeaways

- Pyomo workflow for **intrusive UQ** and **MBDoE** using **ParmEst**, **Pyomo.DoE**
- MBDoE** is similar to **optimal control** with an **information objective** instead of performance
 - General control:** $\phi^* = \arg \min_{\phi \in \Phi} J_{\text{perf}}(\phi)$
 - MBDoE:** $\phi^* = \arg \max_{\phi \in \Phi} \psi(M(\hat{\theta}, \phi))$
- Derivative-consistent sensitivities enable **identifiability** and **model discrimination** in membrane transport
- Closed-loop input design**
 - Estimate $\rightarrow$ Quantify $\rightarrow$ Design $\rightarrow$ Repeat; until target uncertainty is reached

Legend

$\mathbf{x}(t)$: states	$\mathbf{N}(\mathbf{0}, \mathbf{V}_r)$: noise
$\mathbf{u}(t)$: inputs	$\mathbf{V}_r$: measurement covariance
ϕ : operating conditions	$\mathbf{Q}_r$: sensitivity (UQ)
θ : transport parameters	$\mathbf{M}(\hat{\theta})$: FIM
$\hat{\theta}$: parameter estimate	k : sequential experiment index
$\mathbf{y}_r(t)$: measured output, $\mathbf{r}$	Φ : feasible design space

